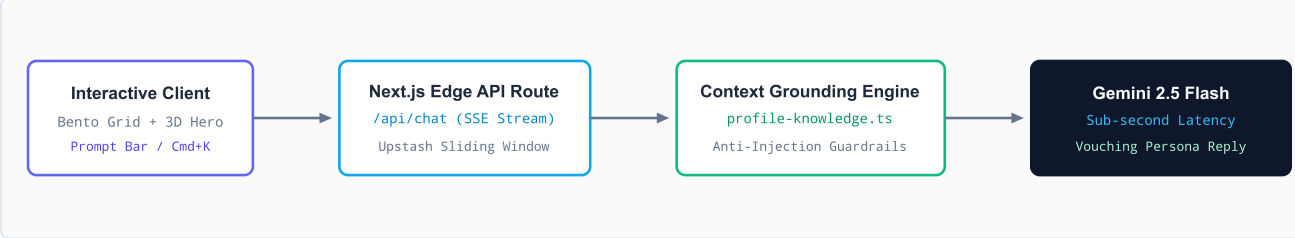
Engineering Architecture & Implementation Blueprint

INTERACTIVE AI PERSONA PORTFOLIO SYSTEM · LINEAR / VERCEL MODERN STANDARD

Target: Autonomous AI Agent
Standard: Production Full-Stack
Revision: 2026.4

1. ARCHITECTURAL INTENT & USER EXPERIENCE PARADIGM

This document specifies the technical design, directory architecture, UI primitives, and security boundaries for deploying an interactive, modern portfolio. The system fuses a dark minimalist aesthetic with high-density Bento grids, an autonomous grounded AI agent persona, and GPU-efficient Three.js WebGL visuals.



2. PRODUCTION TECHNOLOGY STACK

Layer	Package / Technology	Engineering Rationale & Functional Scope
App Framework	`Next.js 15 (App Router)`	React Server Components (RSC), native SEO optimization, zero-latency static rendering.
UI & Tokens	`Tailwind CSS + shadcn/ui`	Design consistency, accessible Radix dialogs/drawers, fluid dark-zinc palette styling.
AI Streaming	`ai + @ai-sdk/google`	Vercel AI SDK handles streaming Server-Sent Events, abort controllers, and hook state.
LLM Runtime	`Google Gemini 2.5 Flash`	Extreme TTFT, accurate strict-grounding execution, and cost-effective serving.
Ambient 3D	`@react-three/fiber + drei`	Procedural shader mesh rendering; runs without heavy external GLTF 3D model downloads.
Abuse Defense	`@upstash/ratelimit + Redis`	Edge-level sliding-window rate limiting (20 req/hr/IP) safeguarding API expenditure.

3. CORE ARCHITECTURAL PRINCIPLES

I. Grounded Advocacy Persona

The agent acts as an optimistic advocate. It highlights technical rigor, measurable impact, and architecture depth. It answers exclusively from verified resume items and refuses unsupported claims.

II. Performance & Zero Lag

No heavy 3D loading screens. Procedural Three.js canvases load asynchronously via `next/dynamic` (SSR: false), ensuring initial DOM paint occurs in under 100ms.

4. DIRECTORY BLUEPRINT & MODULE STRUCTURE

The autonomous agent must construct the project matching this exact file hierarchy:

```
portfolio/ ├── app/ | ├── api/chat/route.ts # Edge runtime API route with Upstash rate-limiting | ├── la
```

5. GROUND TRUTH & PROMPT ENGINEERING PROTOCOL (CONFIG/PROFILE-KNOWLEDGE.TS)

Contains the verified engineering background and behavioral constraints for the persona:

```
export const DEVELOPER_PROFILE = { name: "B S Naveen", title: "AI/ML Systems & Full-Stack Software Eng
```

6. RATE-LIMITED STREAMING ROUTE (APP/API/CHAT/ROUTE.TS)

```
import { streamText } from 'ai'; import { google } from '@ai-sdk/google'; import { ratelimit } from '@
```

7. PROCEDURAL THREE.JS COMPONENT (COMPONENTS/3D/AMBIENTMESH.TSX)

Zero-asset procedural 3D visual featuring pointer reactivity and stateful distortion:

```
'use client'; import { useRef } from 'react'; import { useFrame } from '@react-three/fiber'; import { I
```

8. EXECUTION GUIDE FOR AUTONOMOUS DEVELOPER AGENTS

The developer agent must execute the following sequential installation and verification pipeline:

```
# 1. Project Initialization npx create-next-app@latest portfolio --typescript --tailwind --eslint --app
```

Autonomous Verification & Quality Gate:

1. **Typecheck:** Run `npm run build` to confirm zero TypeScript compilation errors.
2. **Grounding Test:** Send a test query ("Who are you?") to `/api/chat` and assert that the persona self-identifies as Naveen's AI advocate with verified credentials.
3. **Guardrail Test:** Query an unlisted topic ("What is Naveen's home phone?") and verify polite deflection without hallucination.

9. ARCHITECTURAL ACCEPTANCE CHECKLIST

Target Domain	Specification Standard	Validation Criterion
UI Aesthetics	Linear / Vercel Dark Minimalist	Background #09090b, hairline borders, clean typography.
Security	Sliding Window Rate-Limiting	IP limited to 20 req/hr; 429 status code returned on excess.
AI Persona	Confident Grounded Advocate	Strict grounding on `DEVELOPER_PROFILE`; zero hallucinated claims.
3D Graphics	Procedural WebGL Mesh	Zero external 3D asset downloads; locked 60 FPS performance.